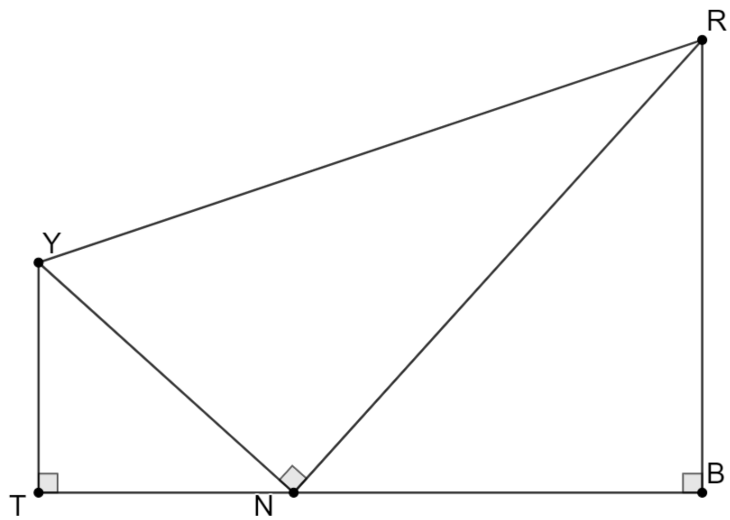
Geometry | Name Answer Key
Unit 7 | Date____________________
Lesson 8 Homework | Period___________________

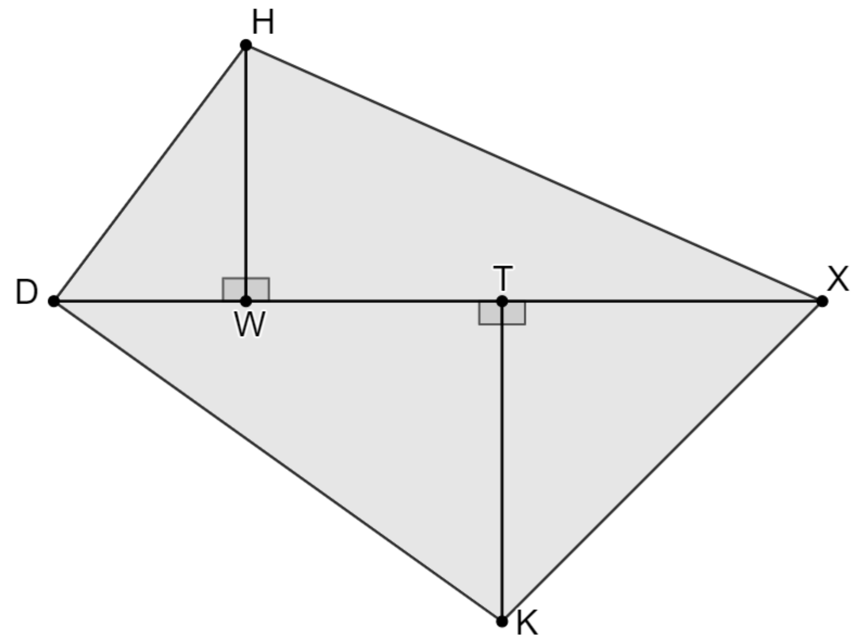
Right triangles YTN, RNY, and NRB are shown where $BR = 6$, $m\angle BNR = 48°$ and $m\angle NRY = 29°$.



Complete the table. In the work column, write the equation or trigonometric ratio used to determine the side measure.

	Side	Work	Side Measure
1.	$\overline{NB}$	$tan\ 48 = \frac{6}{NB}$	5.402
2.	$\overline{NR}$	$sin\ 48 = \frac{6}{NR}$	8.074
3.	$\overline{YR}$	$cos\ 29 = \frac{8.074}{YR}$	9.231
4.	$\overline{YN}$	$tan\ 29 = \frac{YN}{8.074}$	4.475
5.	$\overline{YT}$	$sin\ 42 = \frac{YT}{4.475}$	2.994

Quadrilateral $HXKD$ is composed of right triangles DHW, WHX, TXK, and DTK. $WH = 4$, $TK = 5$, $m\angle WDH = 53°$, $m\angle WHX = 66°$, $m\angle TKX = 45°$, and $m\angle TDK = 35°$.



Complete the table. In the work column, write the equation or trig ratio used to determine the side measure

	Side	Work	Side Measure
6.	$\overline{HX}$	$\cos 66 = \frac{4}{HX}$	9.834
7.	$\overline{XK}$	$\cos\ 45 = \frac{5}{XK}$	7.071
8.	$\overline{DK}$	$\sin 35 = \frac{5}{DK}$	8.717
9.	$\overline{HD}$	$\sin 53 = \frac{4}{HD}$	5.009

10. What is the perimeter of quadrilateral $HXKD$?

30.631